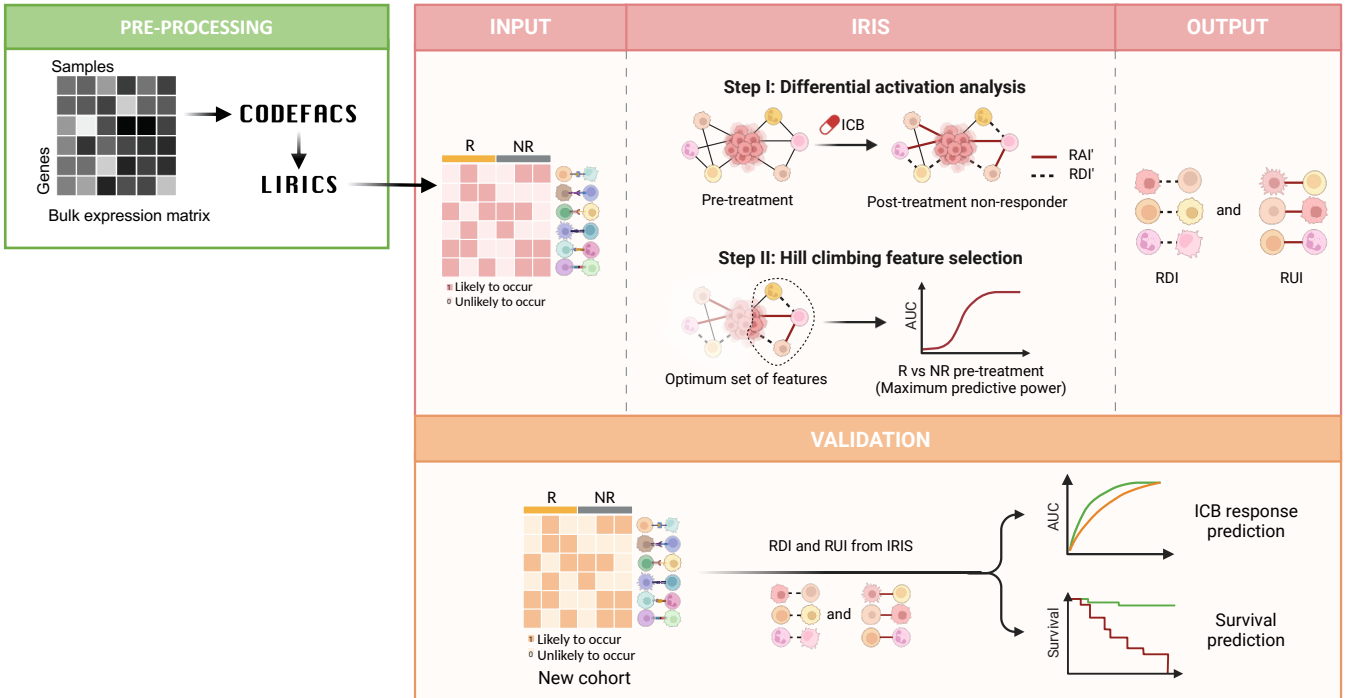
PRE-PROCESSING
Samples
Genes
Bulk expression matrix
CODEFACS
LIRICS
INPUT
R
NR
Likely to occur
Unlikely to occur
IRIS
Step I: Differential activation analysis
ICB
Pre-treatment
Post-treatment non-responder
RAI'
RDI'
Step II: Hill climbing feature selection
Optimum set of features
AUC
R vs NR pre-treatment
(Maximum predictive power)
OUTPUT
and
RDI
RUI
VALIDATION
R
NR
Likely to occur
Unlikely to occur
New cohort
RDI and RUI from IRIS
and
AUC
ICB response prediction
Survival
Survival prediction